Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production experience building distributed backend services, data pipelines, and cloud automation on AWS using Python, Java, and Kubernetes. Current research applies predictive modeling and LLM pipelines to real-time system telemetry.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, C/C++, JavaScript, TypeScript, SQL, Go.

Cloud & Distributed Systems: AWS (S3, EC2, Lambda, SQS), Docker, Kubernetes, ArgoCD, Helm, Terraform, Microservices, REST APIs.

Data & Pipelines: PySpark, pandas, NumPy, PostgreSQL, MySQL, time-series processing, ETL.

Developer Tools: Git, GitLab CI/CD, Jenkins, Linux, Bash, SonarQube, Azure, Google Cloud Platform.

AI & Monitoring: Gemini, Vertex AI, PyTorch, scikit-learn, Grafana, CloudWatch, Datadog.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated production infrastructure workflows with Python (Boto3) across AWS services, replacing legacy scripts and cutting deployment time by **40%** while improving operational reliability.
- Executed **44 production environment upgrades** in 4 months, monitoring live infrastructure, diagnosing anomalies, and resolving issues across Linux, EC2, databases, and distributed services.
- Designed **Ansible-based Blue-Green deployment** framework with automated rollback, reducing production upgrade downtime by **50%** and improving fault recovery.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault, building CI/CD pipelines for reliable, auditable service deployments.
- Served as Quality Coach and SonarQube Specialist, increasing automated test coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built ML pipeline predicting future demand across **263 zones** from **2.3M-row** time-series dataset; two-stage Poisson hurdle model improved MAPE **18.77% to 16.52%** and cut RMSE **43%** over baseline Random Forest, served via real-time REST API.

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).